An Evolutionary Game Theory Analysis on the Rise of the 3-Point Shot in the National Basketball Association

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MATH295: Evolutionary Game Theory

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Abstract

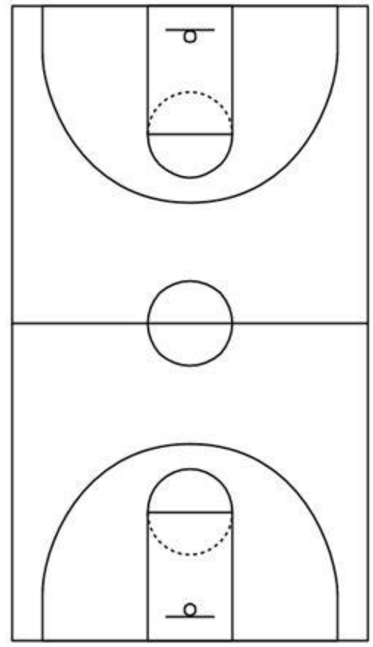
The 3-point shot was introduced to the National Basketball Association (NBA) in 1979, changing the landscape of scoring and style of play (Zając). While initially utilized sparingly, the 3-point shot has seen a dramatic uptick in attempts over the years. Starting during the 2014-15 season, the Golden State Warriors went on a dominant run, appearing in every NBA Finals Series and winning three championships over a five-year stretch. Their success, in large part, was due to their use of the 3-point shot, led by the best 3-point shooter of all time, Stephen Curry. They attempted 3-point shots at a higher rate than most of the league and greatly benefited from doing so. By modeling different shot selection strategies against the rest of the teams, we can discover which are most effective and successful. This paper will use tools of pairwise invasibility to model the success of invasion by theoretical mutants.

Significance Statement

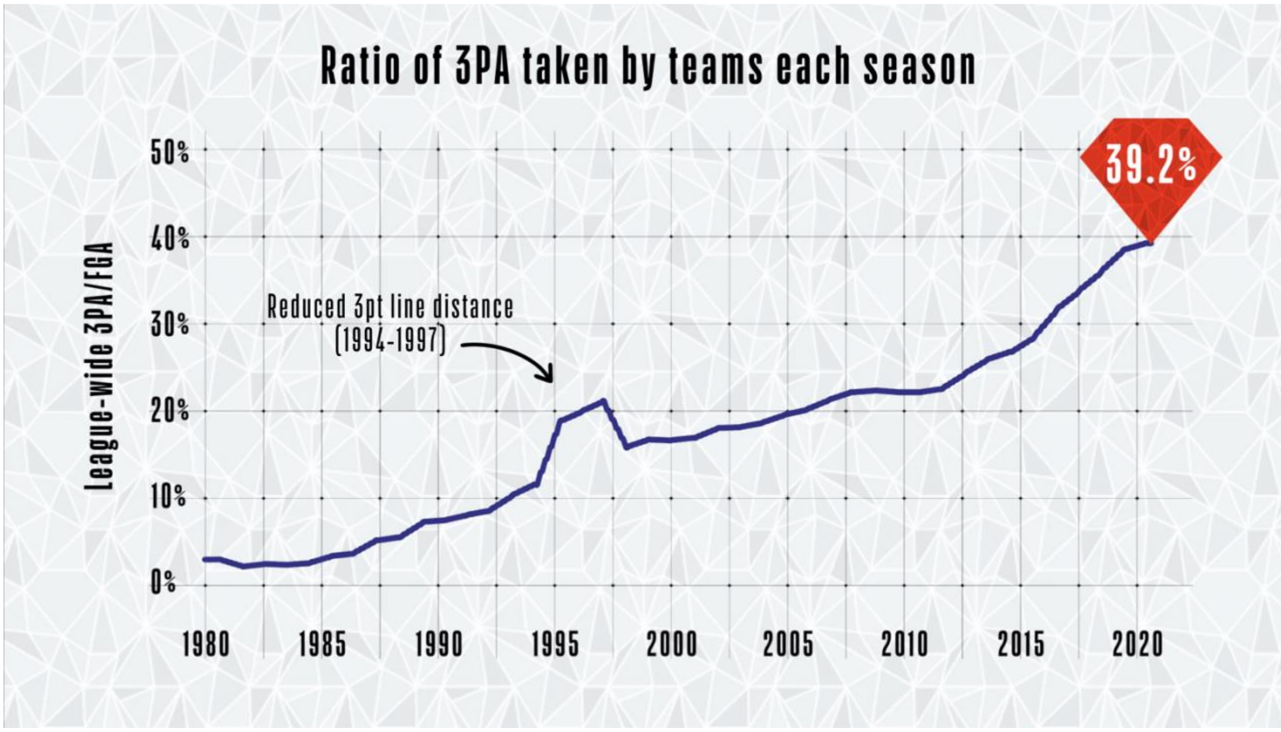
The 3-point shot has become a vital component to an offense's strategy in the NBA and basketball everywhere. The use of the 3-point shot has greatly contributed to teams having record highs in offensive efficiency. How teams decide to capitalize on 3-point shots has a direct link to their success in the league. Using Evolutionary Game Theory, we can attempt to find the best shot selection strategy that will lead to the most offensive success.

Introduction

A 3-point shot is a shot that is attempted beyond the 3-point line which is the arc on the diagram to the right. It is worth 3 points while any shot inside the arc is worth 2 points. Due to the greater distance from the basket, it is usually a more difficult shot to attempt than a 2-point shot, which is why it is awarded a greater value. Thus, depending on how well one can shoot from certain areas of the court, there is a certain risk and reward with shooting 3-point shots. For example, if a team makes 30% of their 3-point attempts, then the expected value of a 3-point shot attempt would be 0.9 points. If they make 50% of their 2-point shot attempts, then the expected value of a 2-point shot attempt would be 1.0 point. For a team with this make-up, shooting more 2-point shots is the best strategy. If the same team made 35% of their 3-point attempts, however, then the expected value of each attempt would climb to 1.05 points. Then shooting more 3-point shots becomes a better strategy. The better teams shoot from beyond the arc, the more payoff they can have with a 3-point shot.


Basketball Court

When the 3-point shot was first introduced in 1979, it was not used much. Multiple factors have contributed to its drastic rise. One reason is 3-point shooting percentages have risen, leading to better payoffs for attempting 3-point shots. Another is revolutionary players and coaches, who have found great success from shooting more 3-point shots. Also, in recent years, more players, regardless of position, have adapted 3-point shooting into their play. The graph below shows how 3-point shots have become more and more popular over the years.



Graph showing the rise in the rate of 3-point attempts in the NBA (Schuhmann)

Basketball can be a very complicated game with many moving parts. In order to make a model, I decided to focus on entire seasons. Trying to model the evolutionary dynamics of individual games or possessions would be quite difficult to do accurately. A season includes 82 regular season games, which gives a sizeable amount of data that can avoid being skewed by outliers. I only looked at the regular season and did not include the playoffs as the intensified environment of the post-season can lead to outlying performances.

To further simplify, I decided to focus on the five-year stretch of the Golden State Warriors' dominant run, starting with the 2014-15 season and ending with the 2018-19 season. This era is very interesting to analyze as the teams show great variance in the frequency at which they attempt 3-point shots, with some as low as 17.9% of field goal attempts and others as high as 51.9% of their field goal attempts, which was done by the Houston Rockets in the 2018-19 season and was the highest frequency of 3-point shots ever. While other eras may have been fruitful to analyze, this five-year stretch seemed most appropriate when looking at the rise of the 3-point shot.

Data and Context

I decided to use basketball-reference.com to gather data for my model as it is a reliable database that contains NBA records and is open access (Zając). I specifically accessed the pages of the 2014-15, 2015-16, 2016-17, 2017-18, and 2018-19 seasons. I used the programming language R to scrape, clean, and manipulate the data so I could use it for my model. The code I used to build the 2014-15 season table can be found here. The code for building the tables for the other seasons is identical save for the website and variable names. The code for how I aggregated the seasons and built the models can be found here.

Description of the Model

To apply the rise of the 3-point shot to evolutionary game theory, I decided to use the tools of pairwise invasibility in my analysis. For pairwise invasibility, there is a resident population and initially rare mutant population. Each has a 1-dimensional strategy. Using the scraped data about each team during the five seasons, I compiled the resident population. The mutant population is hypothetical and arbitrarily chosen by me. The 1-dimensional strategies are each team's shot selection (i.e. frequency of 3-point shot attempts vs frequency of 2-point shot attempts). The fitness is defined as a team's points per game, as that is a measurement of their offensive success.

In an ideal world, I would have used the pairwise invasibility plot function in R, but I was unable to fully understand how to use it. I was, however, able to create a table representative of the resident population. Composed of all 150 teams (30 teams over the 5-year span), the table shows each team's shot selection and success. The headers "ThrPtFreq" and "TwoPtFreq" represent the frequency of 3-point shot attempts and 2-point shot attempts respectively and represent the shot selection. The header "PPG" stands for points per game and represents each team's fitness. The table is organized in descending order, with the fittest teams in the resident population at the top.

Team <chr>	ThrPtFreq <dbl>	TwoPtFreq <dbl>	PPG <dbl>
Milwaukee Bucks 2018-19	0.4193194	0.5806806	118.1
Golden State Warriors 2018-19	0.3830735	0.6158129	117.7
Golden State Warriors 2016-17	0.3582090	0.6406429	115.9
New Orleans Pelicans 2018-19	0.3242950	0.6767896	115.4
Houston Rockets 2016-17	0.4621560	0.5378440	115.3
Philadelphia 76ers 2018-19	0.3424036	0.6575964	115.2
Los Angeles Clippers 2018-19	0.2948571	0.7051429	115.1
Golden State Warriors 2015-16	0.3619702	0.6380298	114.9
Portland Trail Blazers 2018-19	0.3388521	0.6600442	114.7
Oklahoma City Thunder 2018-19	0.3468085	0.6521277	114.5
Toronto Raptors 2018-19	0.3793490	0.6206510	114.4
Sacramento Kings 2018-19	0.3211600	0.6788400	114.2
Washington Wizards 2018-19	0.3695893	0.6304107	114.0
Houston Rockets 2018-19	0.5194508	0.4805492	113.9
Golden State Warriors 2017-18	0.3396005	0.6603995	113.5
Atlanta Hawks 2018-19	0.4030501	0.5969499	113.3
Minnesota Timberwolves 2018-19	0.3143483	0.6845564	112.5
Houston Rockets 2017-18	0.5023753	0.4976247	112.4
Boston Celtics 2018-19	0.3812155	0.6187845	112.4
Brooklyn Nets 2018-19	0.4035674	0.5975474	112.2
Los Angeles Lakers 2018-19	0.3425414	0.6585635	111.8
Denver Nuggets 2016-17	0.3283922	0.6716078	111.7
New Orleans Pelicans 2017-18	0.3193658	0.6806342	111.7
Toronto Raptors 2017-18	0.3775744	0.6224256	111.7
San Antonio Spurs 2018-19	0.2861991	0.7138009	111.7
Utah Jazz 2018-19	0.3935185	0.6064815	111.7
Cleveland Cavaliers 2017-18	0.3785377	0.6202830	110.9
Charlotte Hornets 2018-19	0.3775056	0.6213808	110.7
Denver Nuggets 2018-19	0.3488889	0.6522222	110.7
Cleveland Cavaliers 2016-17	0.3992933	0.6007067	110.3

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Team	ThrPtFreq	TwoPtFreq	PPG
<chr>	<dbl>	<dbl>	<dbl>
Oklahoma City Thunder 2015–16	0.2743056	0.7245370	110.2
Golden State Warriors 2014–15	0.3103448	0.6896552	110.0
Denver Nuggets 2017–18	0.3568129	0.6431871	110.0
Philadelphia 76ers 2017–18	0.3441109	0.6547344	109.8
Minnesota Timberwolves 2017–18	0.2613240	0.7386760	109.5
Washington Wizards 2016–17	0.2850575	0.7160920	109.2
Los Angeles Clippers 2017–18	0.3138173	0.6861827	109.0
Dallas Mavericks 2018–19	0.4211738	0.5776755	108.9
Los Angeles Clippers 2016–17	0.3293269	0.6706731	108.7
Charlotte Hornets 2017–18	0.3137255	0.6851211	108.2
Los Angeles Lakers 2017–18	0.3291855	0.6708145	108.1
Boston Celtics 2016–17	0.3924794	0.6075206	108.0
Indiana Pacers 2018–19	0.2919540	0.7080460	108.0
Portland Trail Blazers 2016–17	0.3217189	0.6782811	107.9
Oklahoma City Thunder 2017–18	0.3450624	0.6549376	107.9
Phoenix Suns 2016–17	0.2553672	0.7446328	107.7
Phoenix Suns 2018–19	0.3352403	0.6647597	107.5
Orlando Magic 2018–19	0.3602694	0.6397306	107.3
Detroit Pistons 2018–19	0.3941110	0.6058890	107.0
Toronto Raptors 2016–17	0.2879147	0.7109005	106.9
Los Angeles Clippers 2014–15	0.3229292	0.6770708	106.7
Sacramento Kings 2015–16	0.2592593	0.7407407	106.6
Oklahoma City Thunder 2016–17	0.2951945	0.7048055	106.6
Brooklyn Nets 2017–18	0.4112903	0.5887097	106.6
Washington Wizards 2017–18	0.3095794	0.6904206	106.6
Houston Rockets 2015–16	0.3700599	0.6299401	106.5
Milwaukee Bucks 2017–18	0.2975904	0.7024096	106.5
Brooklyn Nets 2016–17	0.3708920	0.6291080	105.8
Boston Celtics 2015–16	0.2926009	0.7073991	105.7
Miami Heat 2018–19	0.3681818	0.6318182	105.7

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Team	ThrPtFreq	TwoPtFreq	PPG
<chr>	<dbl>	<dbl>	<dbl>
Minnesota Timberwolves 2016–17	0.2488152	0.7511848	105.6
Indiana Pacers 2017–18	0.2835648	0.7164352	105.6
Portland Trail Blazers 2017–18	0.3229885	0.6758621	105.6
San Antonio Spurs 2016–17	0.2807646	0.7192354	105.3
Dallas Mavericks 2014–15	0.2960373	0.7039627	105.2
Portland Trail Blazers 2015–16	0.3317811	0.6682189	105.1
Indiana Pacers 2016–17	0.2721893	0.7278107	105.1
Charlotte Hornets 2016–17	0.3348946	0.6639344	104.9
Chicago Bulls 2018–19	0.2946530	0.7053470	104.9
Los Angeles Lakers 2016–17	0.2940503	0.7048055	104.6
New York Knicks 2018–19	0.3340883	0.6659117	104.6
Los Angeles Clippers 2015–16	0.3240291	0.6759709	104.5
New York Knicks 2017–18	0.2656784	0.7343216	104.5
Cleveland Cavaliers 2018–19	0.3321918	0.6678082	104.5
Cleveland Cavaliers 2015–16	0.3523810	0.6476190	104.3
New Orleans Pelicans 2016–17	0.3080460	0.6919540	104.3
New York Knicks 2016–17	0.2790960	0.7209040	104.3
Washington Wizards 2015–16	0.2820513	0.7179487	104.1
Utah Jazz 2017–18	0.3570567	0.6429433	104.1
Oklahoma City Thunder 2014–15	0.2615207	0.7384793	104.0
Toronto Raptors 2014–15	0.3013205	0.6986795	104.0
Boston Celtics 2017–18	0.3572268	0.6427732	104.0
Houston Rockets 2014–15	0.3925570	0.6074430	103.9
Phoenix Suns 2017–18	0.3203215	0.6796785	103.9
Detroit Pistons 2017–18	0.3325662	0.6674338	103.8
Milwaukee Bucks 2016–17	0.2893773	0.7106227	103.6
San Antonio Spurs 2015–16	0.2231604	0.7768396	103.5
Memphis Grizzlies 2018–19	0.3424171	0.6587678	103.5
Charlotte Hornets 2015–16	0.3483412	0.6516588	103.4
Atlanta Hawks 2017–18	0.3625731	0.6374269	103.4

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Team <chr>	ThrPtFreq <dbl>	TwoPtFreq <dbl>	PPG <dbl>
Miami Heat 2017-18	0.3587339	0.6424385	103.4
Orlando Magic 2017-18	0.3410943	0.6577416	103.4
San Antonio Spurs 2014-15	0.2691388	0.7308612	103.2
Atlanta Hawks 2016-17	0.3092417	0.6907583	103.2
Miami Heat 2016-17	0.3146853	0.6853147	103.2
Cleveland Cavaliers 2014-15	0.3345499	0.6654501	103.1
Chicago Bulls 2016-17	0.2560276	0.7439724	102.9
Chicago Bulls 2017-18	0.3502252	0.6509009	102.9
Portland Trail Blazers 2014-15	0.3162791	0.6837209	102.8
Atlanta Hawks 2015-16	0.3364929	0.6646919	102.8
Sacramento Kings 2016-17	0.2911084	0.7088916	102.8
New Orleans Pelicans 2015-16	0.2770664	0.7229336	102.7
Toronto Raptors 2015-16	0.2878229	0.7134071	102.7
San Antonio Spurs 2017-18	0.2822014	0.7166276	102.7
Atlanta Hawks 2014-15	0.3206854	0.6793146	102.5
Phoenix Suns 2014-15	0.2913753	0.7097902	102.4
Minnesota Timberwolves 2015-16	0.2017220	0.7982780	102.4
Philadelphia 76ers 2016-17	0.3493552	0.6506448	102.4
Dallas Mavericks 2015-16	0.3400713	0.6611177	102.3
Dallas Mavericks 2017-18	0.3818393	0.6181607	102.3
Indiana Pacers 2015-16	0.2699531	0.7288732	102.2
Orlando Magic 2015-16	0.2557604	0.7453917	102.1
Detroit Pistons 2015-16	0.3032407	0.6967593	102.0
Denver Nuggets 2015-16	0.2775176	0.7224824	101.9
Chicago Bulls 2015-16	0.2448513	0.7562929	101.6
Denver Nuggets 2014-15	0.2840779	0.7159221	101.5
Boston Celtics 2014-15	0.2798635	0.7201365	101.4
Sacramento Kings 2014-15	0.2044610	0.7955390	101.3
Detroit Pistons 2016-17	0.2635135	0.7376126	101.3
Orlando Magic 2016-17	0.3000000	0.7000000	101.1

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Team <chr>	ThrPtFreq <dbl>	TwoPtFreq <dbl>	PPG <dbl>
Phoenix Suns 2015-16	0.3014019	0.6985981	100.9
Chicago Bulls 2014-15	0.2689988	0.7310012	100.8
Utah Jazz 2016-17	0.3270440	0.6729560	100.7
Memphis Grizzlies 2016-17	0.3169856	0.6830144	100.5
Miami Heat 2015-16	0.2203182	0.7784578	100.0
New Orleans Pelicans 2014-15	0.2328106	0.7671894	99.4
Memphis Grizzlies 2017-18	0.3164251	0.6823671	99.3
Memphis Grizzlies 2015-16	0.2212919	0.7787081	99.1
Milwaukee Bucks 2015-16	0.1897810	0.8102190	99.0
Sacramento Kings 2017-18	0.2787456	0.7212544	98.8
Brooklyn Nets 2015-16	0.2180095	0.7819905	98.6
Detroit Pistons 2014-15	0.2902098	0.7097902	98.5
Los Angeles Lakers 2014-15	0.2207944	0.7803738	98.5
Washington Wizards 2014-15	0.2028986	0.7971014	98.5
New York Knicks 2015-16	0.2559524	0.7440476	98.4
Memphis Grizzlies 2014-15	0.1842424	0.8157576	98.3
Brooklyn Nets 2014-15	0.2397590	0.7602410	98.0
Dallas Mavericks 2016-17	0.3669502	0.6342649	97.9
Milwaukee Bucks 2014-15	0.2231707	0.7768293	97.8
Minnesota Timberwolves 2014-15	0.1790865	0.8209135	97.8
Utah Jazz 2015-16	0.2972637	0.7027363	97.7
Philadelphia 76ers 2015-16	0.3273810	0.6726190	97.4
Indiana Pacers 2014-15	0.2548077	0.7451923	97.3
Los Angeles Lakers 2015-16	0.2900943	0.7099057	97.3
Orlando Magic 2014-15	0.2355072	0.7644928	95.7
Utah Jazz 2014-15	0.2739899	0.7260101	95.1
Miami Heat 2014-15	0.2616580	0.7383420	94.7
Charlotte Hornets 2014-15	0.2260355	0.7739645	94.2
Philadelphia 76ers 2014-15	0.3184019	0.6815981	92.0
New York Knicks 2014-15	0.2402439	0.7597561	91.9

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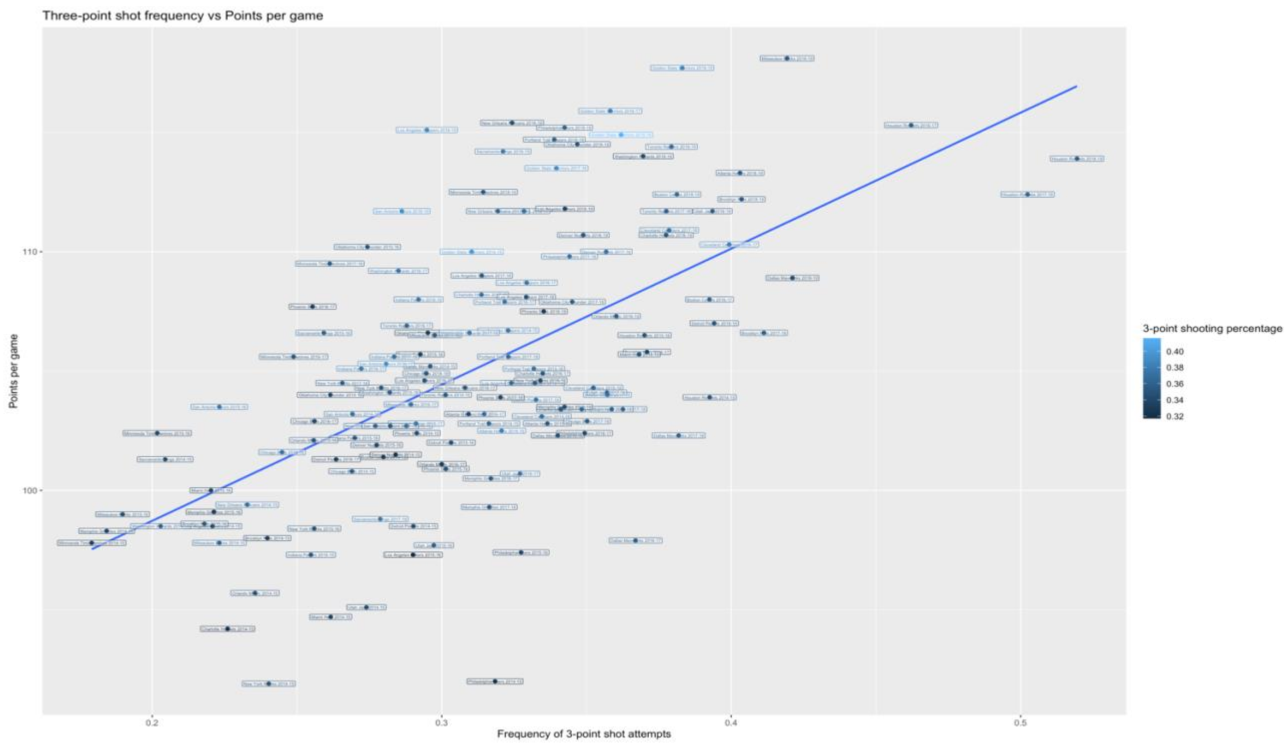
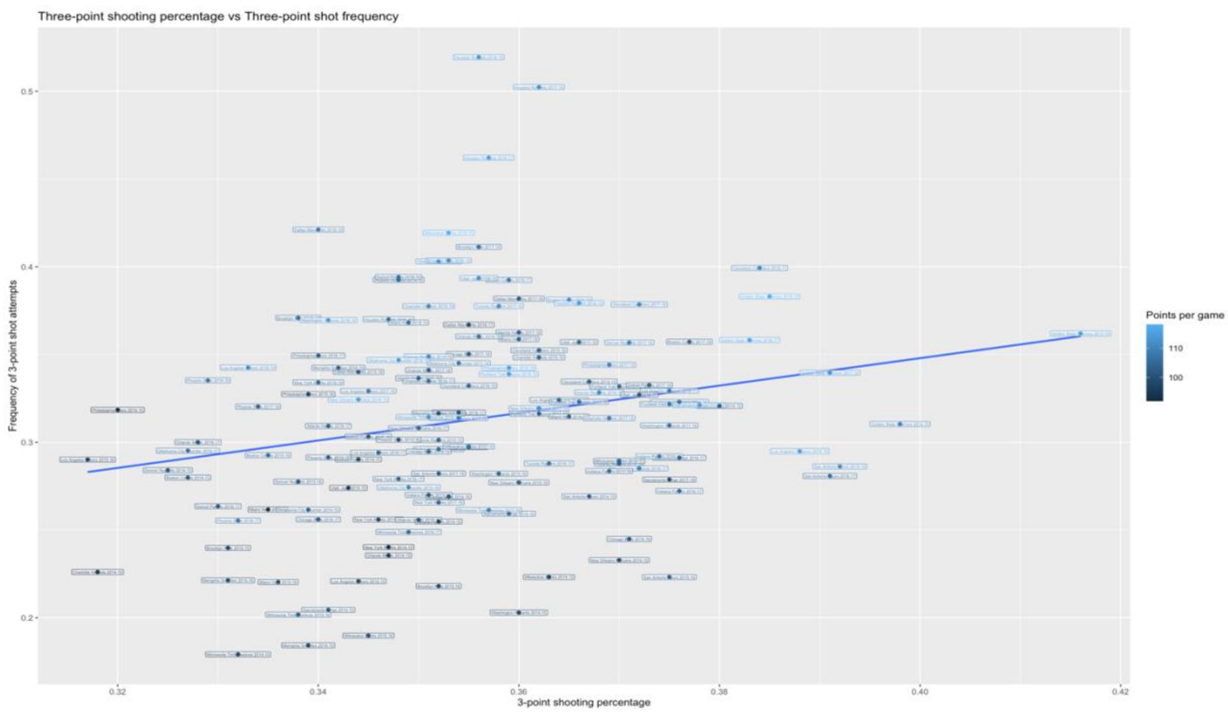
I also created statistical models to demonstrate the relationship between 3-point frequency, points per game, and 3-point shooting percentage (i.e. how well teams shot 3-

pointers). While potentially difficult to make out, each point is labelled with the team it represents. This first model displays the relationship between the frequency of 3-point shots and points per game, with 3-point shooting percentage coloring the dots for additional context. From this scatterplot, you can see a strong, positive correlation between the frequency of 3-point shot attempts and points per game.



This next model displays the relationship between 3-point shooting percentage and 3-point shot frequency, with points per game coloring the dots. The purpose of this plot was to see if how well teams shot the ball affected how much they wanted to shoot it. From this scatterplot, you can see a moderate and slightly positive relationship between 3-point shooting percentage and the frequency of 3-point shot attempts.



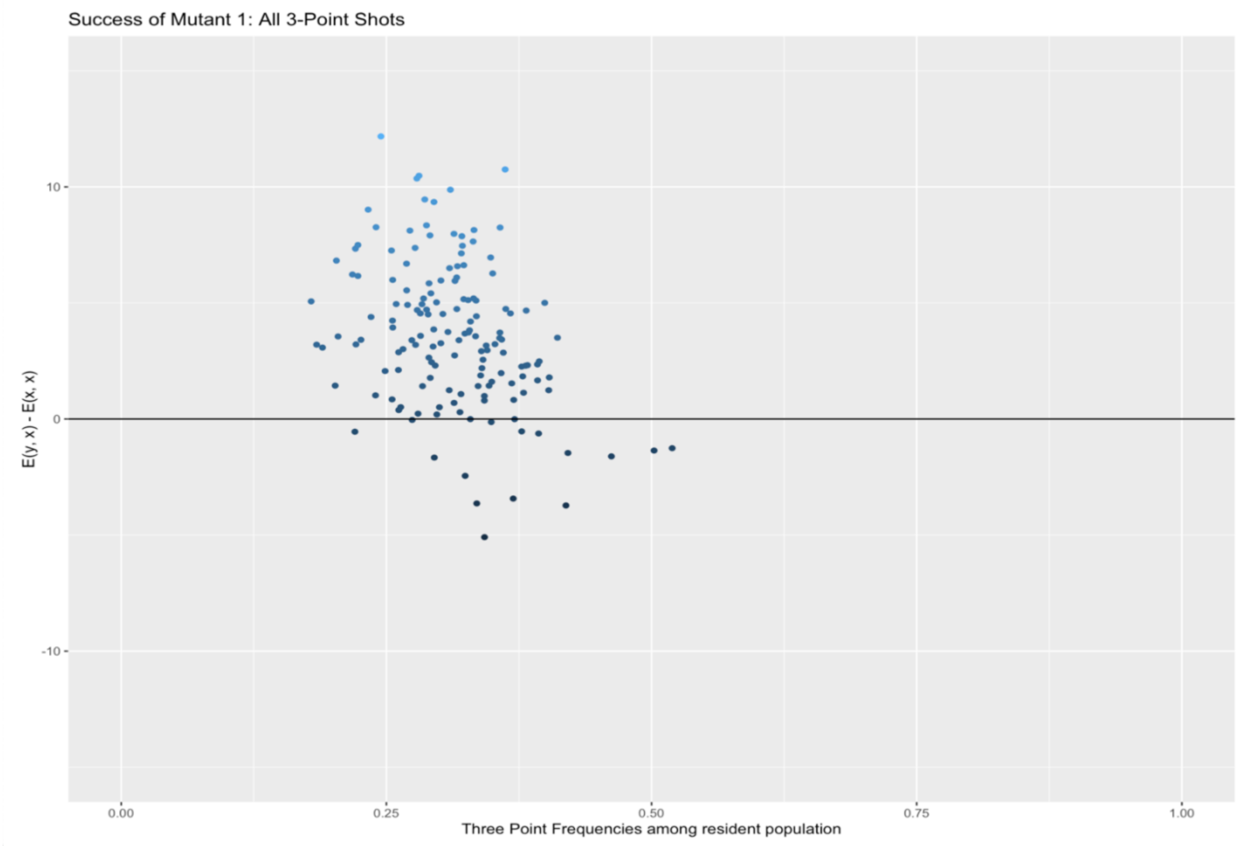
To create models related to pairwise invasibility, I chose mutants to test against the resident population, calculated their fitness, and compared their fitness against the resident population. To calculate a mutant's fitness, I used the following equation:

$$\Big(\underbrace{(\mathrm{FGA}\cdot 3\mathrm{pt\ freq})}_{\text{3pt shot attempts}}\cdot\underbrace{3\mathrm{pt}\ \%}_{\text{shooting percentage}}\cdot\underbrace{3}_{\text{3pt shot value}}\Big)+\Big(\underbrace{(\mathrm{FGA}\cdot 2\mathrm{pt\ freq})}_{\text{2pt shot attempts}}\cdot\underbrace{2\mathrm{pt}\ \%}_{\text{shooting percentage}}\cdot\underbrace{2}_{\text{2pt shot value}}\Big)+\underbrace{\mathrm{FT}}_{\text{free throws}}$$

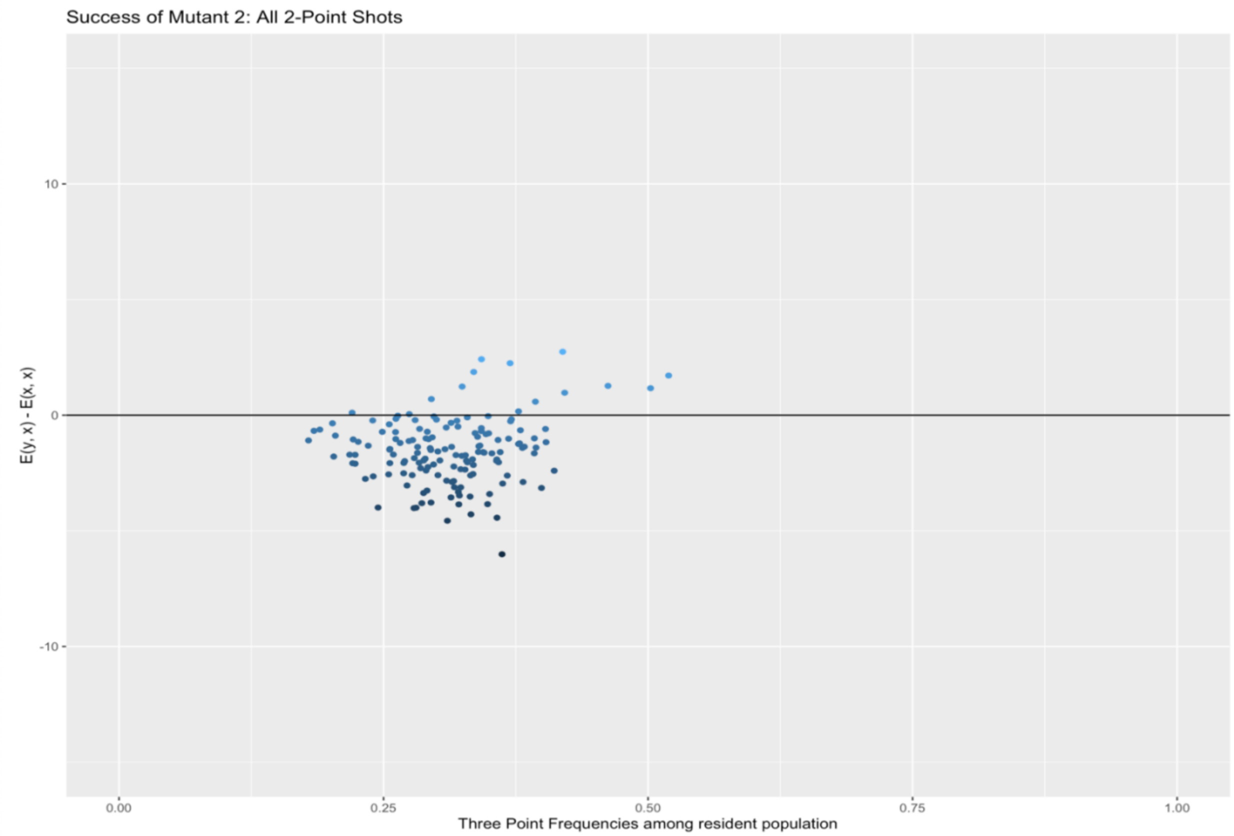
where "FGA" represents the average number of total field goal attempts per game, "3pt freq" represents the average frequency of 3-point shot attempts per game, "3pt%" represents the average 3-point shooting percentage per game, "2pt freq" represents the average frequency of 2-point shot attempts per game, "2pt%" represents the average 2-point shooting percentage per game, and "FT" represents the average number of free throw points per game. Free throws are included as they contribute to the total points per game for a team.

To compare the fitness of the mutant population vs the fitness of the resident population, I simply calculated *E(y, x) – E(x, x)*, where *E(y, x)* is the mutant's fitness and *E(x, x)* is the resident's fitness. I chose four hypothetical mutants to compete against the resident teams: all 3-point shots, all 2-point shots, half 3-point and half 2-point shots, and the shot selection of the team with the most points per game in the 2023-24 season, the Indiana Pacers.

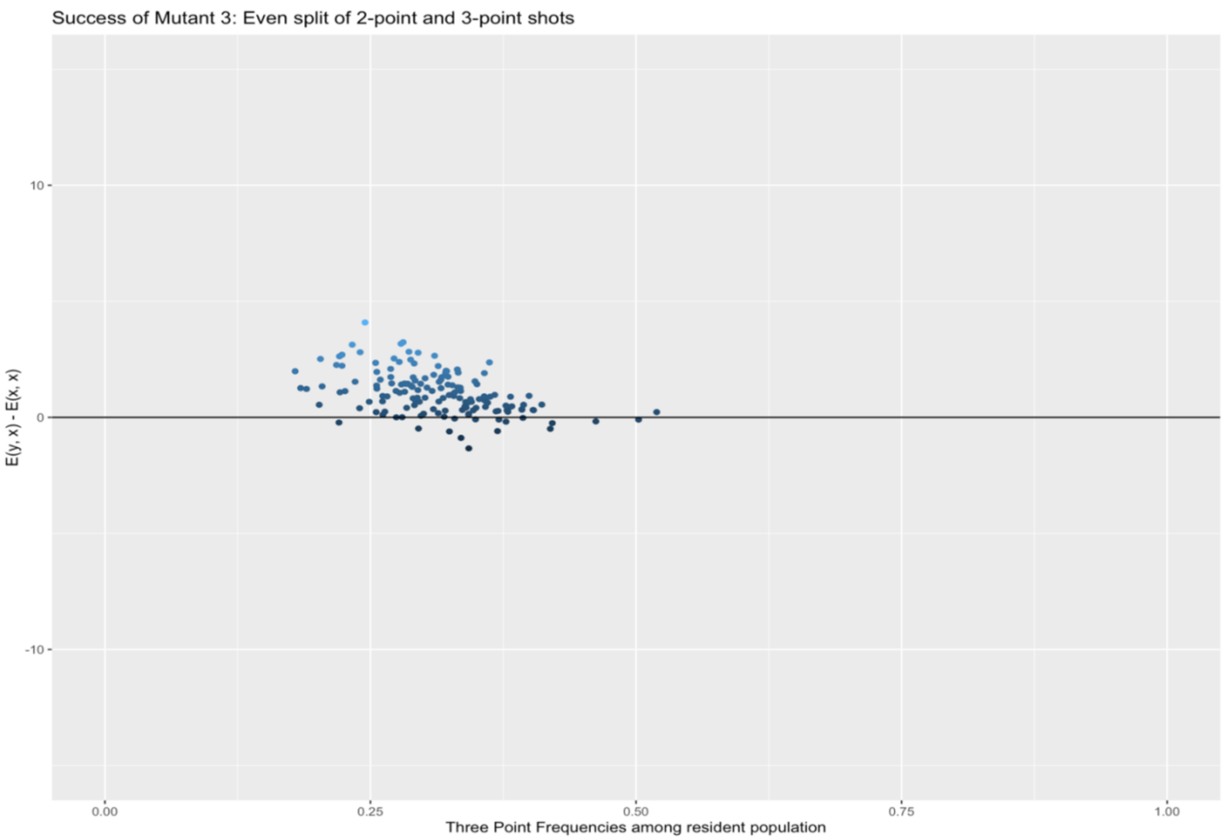
Below is a visual representation of this comparison for each mutant.



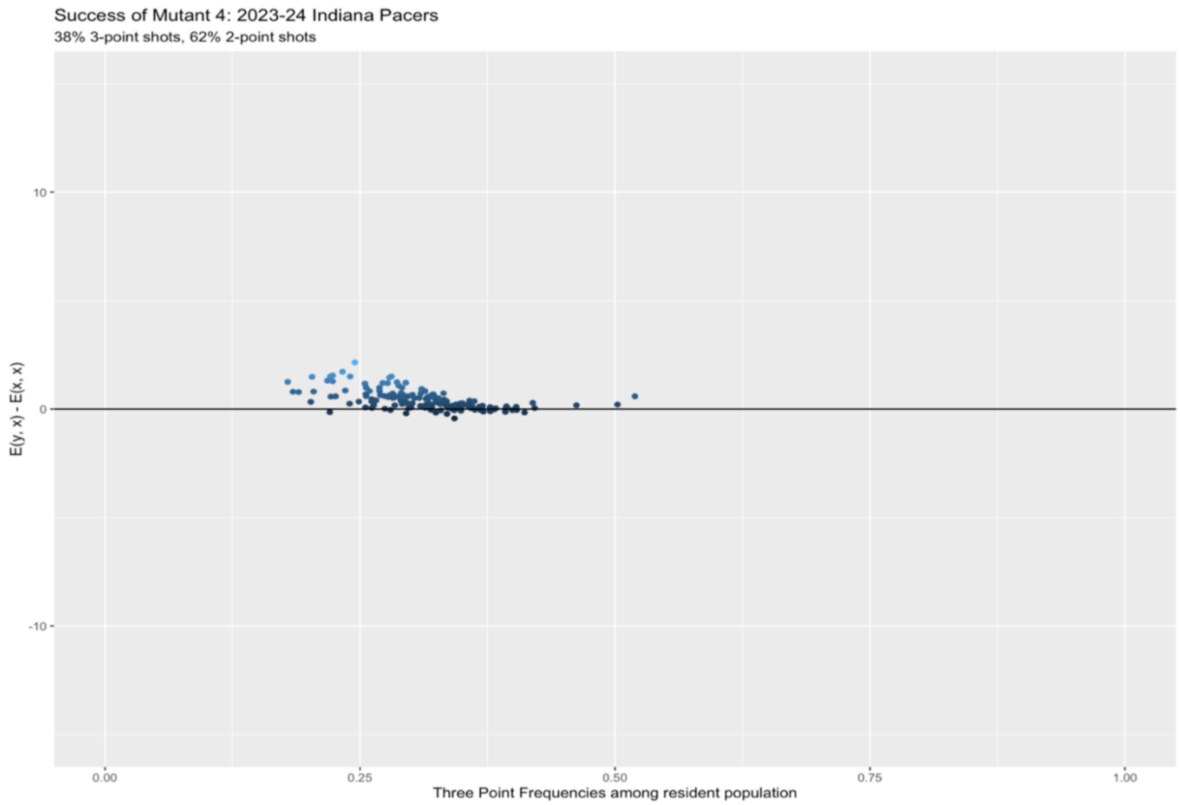
With a shot selection composed of 100% 3-point shot frequency and 0% 2-point shot frequency, this mutant had better fitness against most of the resident population. The difference of *E(y, x) – E(x, x)* ranged from -5.0905 to 12.1762, with a median of 3.4549 and a mean of 3.634916.



With a shot selection composed of 0% 3-point shot frequency and 100% 2-point shot frequency, this mutant had a worse fitness against most of the resident population. The difference *E(y, x) – E(x, x)* ranged from -6.0112 to 2.743, with a median of -1.5773 and mean of -1.539545.

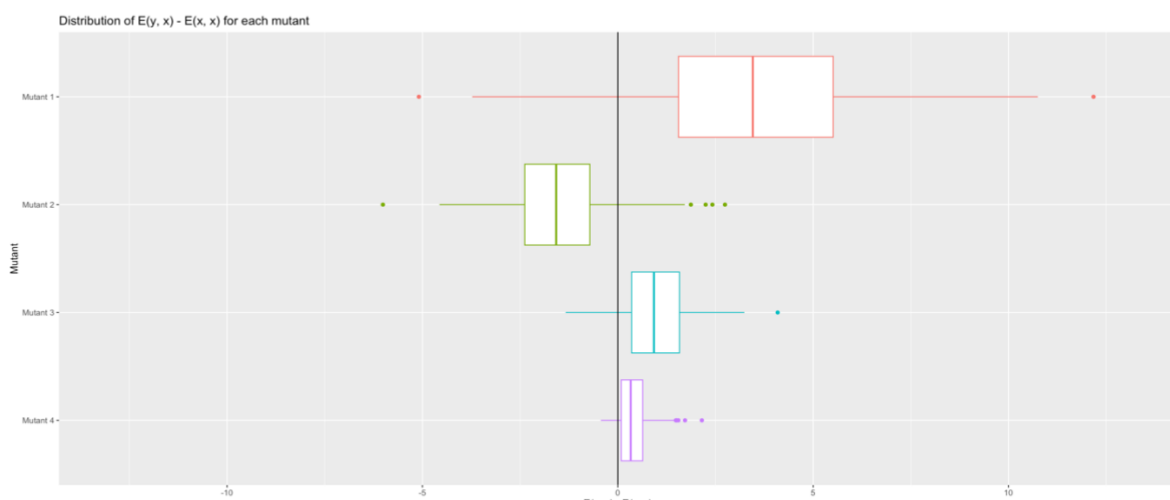


With a shot selection composed of 50% 3-point shot frequency and 50% 2-point shot frequency, this mutant had a better fitness than most of the resident population. The difference *E(y, x) – E(x, x)* ranged from -1.33475 to 4.0917 with a median of 0.92465 and a mean of 1.047685.



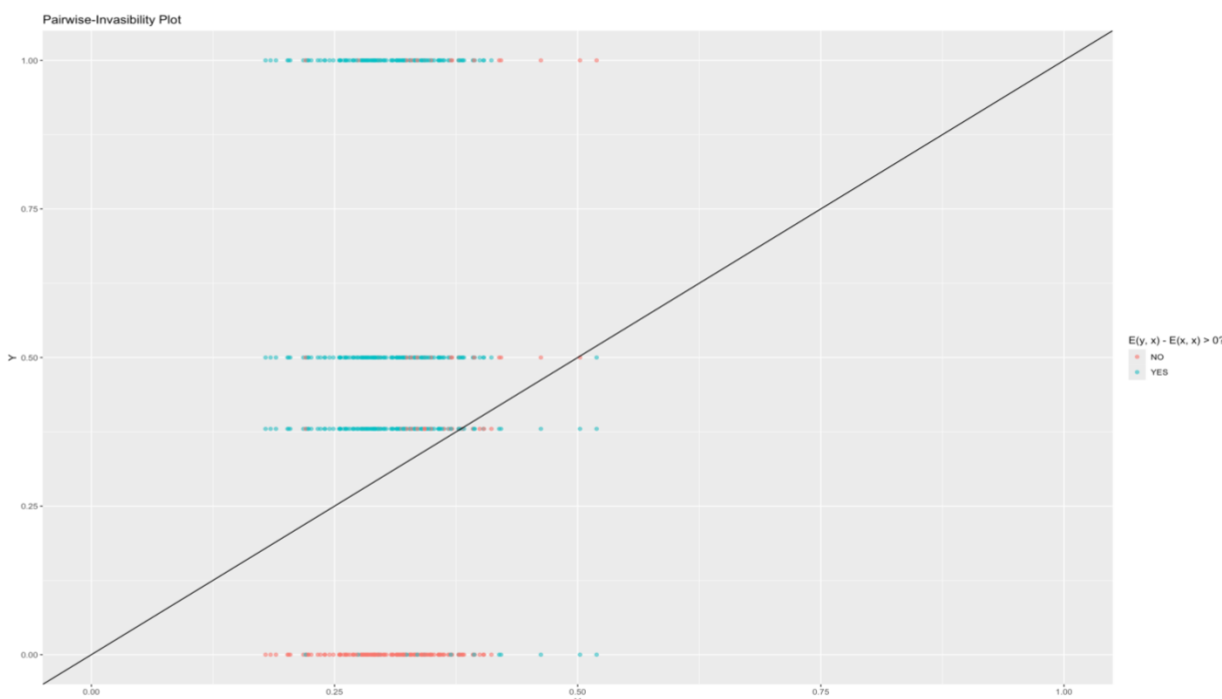
With a shot selection of 38% 3-point shot frequency and 62% 2-point shot frequency, this mutant had a better fitness than most of the resident population. The difference *E(y, x) – E(x, x)* ranged from -0.43337 to 2.15142 with a median of 0.330234 and a mean of 0.42675. This mutant's failures were very close to being successes.

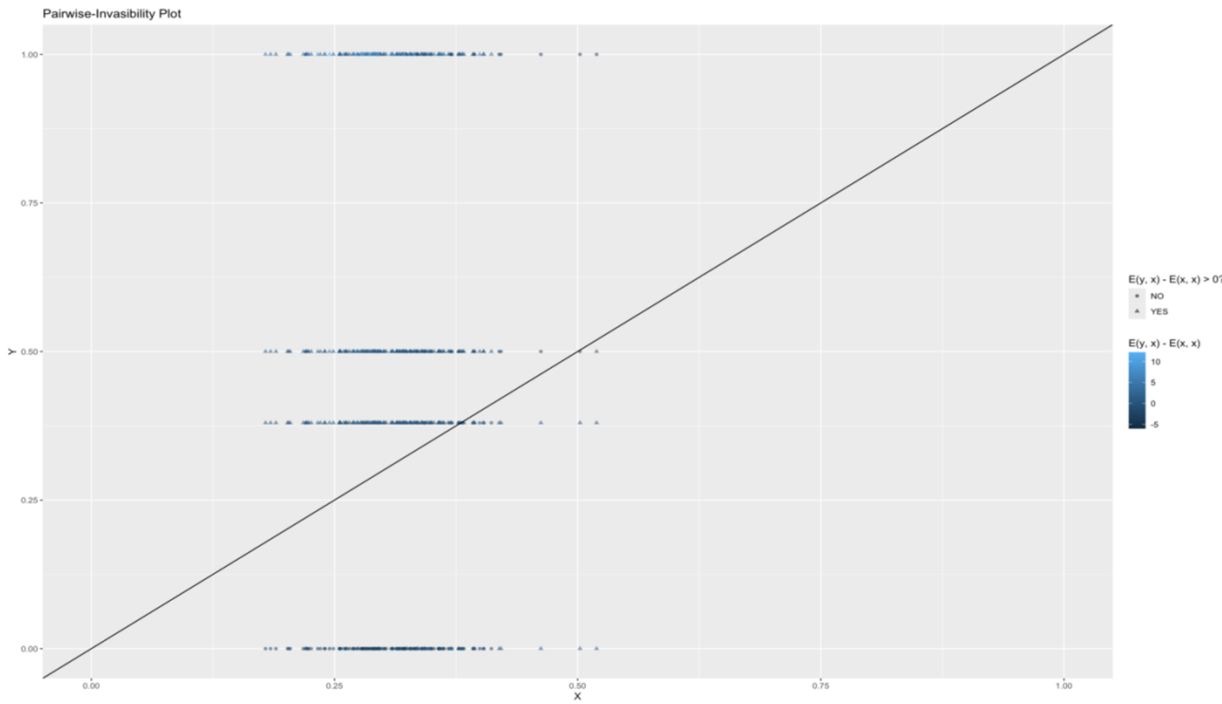
To compare the distributions of *E(y, x) – E(x, x)* for each mutant, I created this boxplot:



From this boxplot, it can be seen that Mutant 1, Mutant 3, and Mutant 4 all performed well against the resident population while Mutant 2 performed very poorly. Mutant 1 had the widest range, finding both the most success and some of the worst failure.

The following models are pairwise-invasibility plots I created. Since the data is discrete, they look different than traditional pairwise-invasibility plots but can still demonstrate the success of mutant invasion. The models are the same save for how the success of the mutant is visualized. In the first, it is simply by color, while in the second, it is by shape and color to show how successful each point was. The points are grouped into four distinct areas, representative of the four mutants. From top to bottom, it is Mutant 1, Mutant 3, Mutant 4, and Mutant 2.





As similarly seen with the previous models, Mutants 1, 3, and 4 found lots of success while Model 2 found lots of failure.

Results & Analysis of the Model

From the models shown above, it seems as though shooting 3-point shots at a higher rate allowed the hypothetical mutant to successfully invade the resident population. However, the wide range of Mutant 1 does show that this strategy can be very unsuccessful for some teams. Since the data is discrete, it is difficult to make out the full patterns of the model and determine stability points. Since shot selection has seen continual change since the inception of the 3-point shot, I do not believe that teams have found an evolutionary stable strategy (i.e. one that is immune to invasion). However, as teams have increasingly shot 3-pointers over the years, I could imagine a convergent stable strategy towards a higher frequency of 3-point shots.

Any strategy that comes about has its limits as teams and playing styles change as well as their skill levels in different aspects of basketball. These inevitable changes make it difficult for a strategy to be evolutionarily stable and difficult for a true convergent stable strategy to exist, but not impossible.

Discussion and Next Steps

While the model shows high favorability towards strategies with high 3-point shot frequencies, it is entirely unrealistic for a team to shoot only 3-point shots in a game. It makes the offense far too predictable for defenses to play against, providing an advantage for their opponent. It could also be challenging for players to maintain good shooting percentages for an entire season if they were shooting such a high volume of 3-point shots a game.

The model I created is a very oversimplified version of basketball that only focuses on shot attempts. It does not account for how free throw rates may change due to varying shot selection. A team that only shoots 3-point shots is less likely to be fouled and be awarded free throws as those types of shots do not usually draw contact worthy of a foul. In a more accurate model, it would be useful to account for the effects of shot selection on free throw rate as it could have some impact on a team’s points per game.

The model also does not account for defensive strategies employed by opposing teams. Defenses can take on numerous forms. Teams vary in which parts of the offense they intend to cut off (and in turn which they are willing to give up), how tightly and aggressively they defend opposing players, whether they utilize a man-to-man or zone scheme, and more. In the future, I would explore how to analyze different defensive strategies and their success against different offensive strategies.

Another way I could have altered my model was choosing a different measurement of success and fitness. I chose to use points per game but could have opted for offensive efficiency. This can be measured by the number of points per possession a team scores, looking at how an offense optimizes each offensive opportunity over the course of a game, instead of the result of a game.

If time had allowed, I would have loved to choose more mutants to test to give my model more data. This could have allowed for more patterns to emerge. Also, I think it would have been intriguing to look at the 1979-80 season, when the 3-point shot was introduced. In that time, there is a truly new mutant entering the population. The 2023-24 season would have been interesting to analyze as well as the two teams with the best winning records had the highest 3-point shot frequency and lowest 3-point shot frequency, respectively. It would be interesting to investigate potential coexistence of strategies within the population. In an ideal world, I would have loved to find the true best shot selection for a given team. I hope to see more analysis of basketball with evolutionary game theory in the future.

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